10.5.4 Analytical Ultra centrifugation:

Analytical ultracentrifugation, developed by Svedberg, is a versatile method for the separation of small particles such as particles of 10 microns in size. It can operate safely at about rotor speeds upto 60,000 – 10,00,000 rpm. Protein molecules, under the influence of high centrifugal force can be made to sediment toward one end of a centrifuge tube. The rate of sedimentation for a protein under a certain centrifugal force depends on the density, shape and size of the molecule. It is used to determine the sedimentation co-efficient and molecular weight of the macromolecules in solution.

10.6 Spectrophotometry

The use of spectroscopic methods to evaluate bio-molecules continues to be an important area in Biochemistry. Spectroscopy is the study of the absorption and emission of radiation by matter. Using spectroscopic methods, it is possible to analyze the colour developed by the bio-molecule exhibited by the substances that absorb radiation from the visible region of the spectrum. Measurements of the intensity and wavelength of the radiation that is either absorbed or emitted provide the basis for sensitive method of detection and quantification. The study of functions of the body in both health and diseased states require the analysis of blood, serum and body fluids. The applications of spectrometry has produced more diagnostic kits to quantify the constituents present in blood, tissue, urine and other biological material.

We might have noticed that the colours of two solutions of the same substance, one a deep and another lighter colour. When the colour of the solution deepens, it is an indication that it is more concentrated. This forms the basis of spectrophotometry, where the intensity of the colour is a direct measure of the amount of the material present in the solution. Light is a form of electromagnetic radiation. When it falls on a solution one can expect three changes. a). the light can be reflected by the substance b). the light can be absorbed by the substance c). only certain wavelengths can be absorbed and the remaining transmitted. The absorbance of transmitted light is of prime importance in spectrophotometry. The observed colour is due to the absorption of a certain wavelength of light in the visible spectrum and transmittance of the other wavelength.

Photometric instruments are designed to measure the intensity of radiation. Instruments prefixed with the term 'spectro' are capable of splitting radiation into a spectrum

10.6.1 Principle of Beer-Lambert's law

The absorption of light by a solution is shown in Fig.10.15. When a beam of monochromatic light (I_o) is allowed to pass through a homogenous light absorbing

Unit 10.indd 274 21-05-2018 15:13:14